

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester III Examination 2025-26 (NEP)

DST-Centre for Interdisciplinary Mathematical Sciences

Skill Enhancement Course (SEC)**Paper Code: CIMSE31 : Python Programming for Data Analytics**

Time: 2:30 Hours

Maximum Marks: 70

*Instruction: Write your Roll No. at the top immediately on the receipt of this question paper.**Note: Answer any five questions including question 1, which is compulsory. Each question carries equal marks (14 marks each).***Question 1.** Short Answer Questions (Compulsory): [14]

(a) Which keyword is used to define a function in Python? [2]

(b) What will be the output produced by the following code? [2]

```
a = 5
b = -3
c = 25
d = -10
print(a + b + c > a + c - b * d)
print(str(a + b + c > a + c - b * d) == 'True')
print(len(str(a + b + c > a + c - b * d)) == len(str(bool(1))))
```

(c) Which data structure stores data in key-value pairs? [2]

(d) What symbol is used for comments in Python? [2]

(e) What will be the output produced by the following code? [2]

```
a = 3 + 5 / 8
b = int(3 + 5 / 8)
c = 3 + float(5 / 8)
d = 3 + float(5) / 8
e = 3 + 5.0 / 8
f = int(3 + 5 / 8.0)
print(a, b, c, d, e, f)
```

(f) What type of plot is used to show the relationship between two variables? [2]

(g) What is the basic unit of a Pandas DataFrame column? [2]

Question 2.

(a) What is a function in Python? Explain how functions are defined and invoked. Discuss different types of function arguments with examples. [14]

OR

- (a) Explain the concept of variable scope in Python. Distinguish between local, global, and nonlocal variables with suitable examples. How does Python manage the lifetime of variables? [14]
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Question 3.

- (a) Differentiate between recursive and iterative approaches with examples. [14]

OR

- (a) Write and explain a Python program to illustrate the difference between **break** and **continue** statements. [14]
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Question 4.

- (a) What are tuples in Python? Compare tuples with lists in terms of mutability, performance, and use cases. [14]

OR

- (a) Explain the concept of sets in Python. Discuss different set operations such as union, intersection, difference, and symmetric difference with suitable examples. Highlight the importance of sets in eliminating duplicate elements. [14]
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Question 5.

- (a) What is a Pandas Series? Explain its structure, creation, indexing, and basic operations with suitable examples. [14]

OR

- (a) Describe the structure of a Pandas DataFrame. Explain different ways of creating DataFrames and performing basic operations such as selection, filtering, and aggregation. [14]
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Question 6.

- (a) Explain the importance of data visualization in analytics. Describe the structure of Matplotlib and its basic workflow for creating plots. [14]

OR

- (a) Explain how line plots are created using Matplotlib. Discuss customization options such as labels, titles, legends, and grids with examples. [14]
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Question 7.

- (a) Explain data aggregation in Pandas. Discuss the use of `groupby()` for aggregation and summarization. Illustrate common aggregation functions such as `sum`, `mean`, `count`, and `agg()` with examples. [14]

OR

- (a) Explain the process of data cleaning in Pandas. Describe different methods available in Pandas for detecting and handling missing data. [14]